ETALE BASE CHANGE OF A FINITE LOCALLY FREE GROUPOID: INVARIANTS ETALE AND U' = M' X_M U

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file groupoids.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: this chapter's pool has 47 of 128 labels already delivered by the S7 line (S7-gp-* registrations, statement_anchor-prose scan; withdrawn cards excluded); none of the 11 accepted labels of this batch is among them (gp-occupied.json, 2026-10-14).

Source reference. Original lemma 03BJ: sources/stacks-chow/tags.txt, source lines 4350–4350. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03BK: sources/stacks-chow/tags.txt, source lines 4351–4351. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03BL: sources/stacks-chow/tags.txt, source lines 4352–4352. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01Z4: sources/stacks-chow/tags.txt, source lines 2579–2579. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 081E: sources/stacks-chow/tags.txt, source lines 10214–10214. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0476: sources/stacks-chow/tags.txt, source lines 5497–5497. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00GA: sources/stacks-chow/tags.txt, source lines 600–600. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 09XK: sources/stacks-chow/tags.txt, source lines 12533–12533. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 09XJ: sources/stacks-chow/tags.txt, source lines 12532–12532. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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Source reference. Original reference 09XI: sources/stacks-chow/tags.txt, source lines 12531–12531. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02GW: sources/stacks-chow/tags.txt, source lines 3243–3243. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02LC: sources/stacks-chow/tags.txt, source lines 3403–3403. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $f : (U', R', s', t') \to (U, R, s, t, c)$ be a morphism of groupoid schemes over S.*

(1) *U, R, U', R' are affine,*
(2) *s, t, s', t' are finite locally free,*
(3) *the diagrams*

$$\begin{array}{ccc} R' & \xrightarrow{f} & R \\ {\scriptstyle s'}\downarrow & & \downarrow{\scriptstyle s} \\ U' & \xrightarrow{f} & U \end{array} \qquad \begin{array}{ccc} R' & \xrightarrow{f} & R \\ {\scriptstyle t'}\downarrow & & \downarrow{\scriptstyle t} \\ U' & \xrightarrow{f} & U \end{array} \qquad \begin{array}{ccc} G' & \xrightarrow{f} & G \\ \downarrow & & \downarrow \\ U' & \xrightarrow{f} & U \end{array}$$

are cartesian where G and G' are the stabilizer group schemes, and

(4) *$f : U' \to U$ is étale.*

Then the map $C \to C'$ from the R-invariant functions on U to the R'-invariant functions on U' is étale and $U' = \mathrm{Spec}(C') \times_{\mathrm{Spec}(C)} U$.

Proof. Set $M = \mathrm{Spec}(C)$ and $M' = \mathrm{Spec}(C')$. Write $U = \mathrm{Spec}(A)$, $U' = \mathrm{Spec}(A')$, $R = \mathrm{Spec}(B)$, and $R' = \mathrm{Spec}(B')$. We will use the results of Lemmas 03BJ, 03BK, and 03BL without further mention.

Assume C is a strictly henselian local ring. Let $p \in M$ be the closed point and let $p' \in M'$ map to p. Claim: in this case there is a disjoint union decomposition $(U', R', s', t', c') = (U, R, s, t, c) \amalg (U'', R'', s'', t'', c'')$ over (U, R, s, t, c) such that for the corresponding disjoint union decomposition $M' = M \amalg M''$ over M the point p' corresponds to $p \in M$.

The claim implies the lemma. Suppose that $M_1 \to M$ is a flat morphism of affine schemes. Then we can base change everything to M_1 without affecting the hypotheses (1) – (4). From Lemma 03BK we see M_1, resp. M_1' is the spectrum of the R_1-invariant functions on U_1, resp. the R_1'-invariant functions on U_1'. Suppose that $p' \in M'$ maps to $p \in M$. Let M_1 be the spectrum of the strict henselization of $\mathcal{O}_{M,p}$ with closed point $p_1 \in M_1$. Choose a point $p_1' \in M_1'$ mapping to p_1 and p'. From the claim we get

$$(U_1', R_1', s_1', t_1', c_1') = (U_1, R_1, s_1, t_1, c_1) \amalg (U_1'', R_1'', s_1'', t_1'', c_1'')$$

and correspondingly $M_1' = M_1 \amalg M_1''$ as a scheme over M_1. Write $M_1 = \mathrm{Spec}(C_1)$ and write $C_1 = \mathrm{colim}\, C_i$ as a filtered colimit of étale C-algebras. Set $M_i = \mathrm{Spec}(C_i)$. The $M_1 = \lim M_i$ and similarly for the other schemes. By Limits, Lemmas 01Z4 and 081E we can find an i such that

$$(U_i', R_i', s_i', t_i', c_i') = (U_i, R_i, s_i, t_i, c_i) \amalg (U_i'', R_i'', s_i'', t_i'', c_i'')$$

We conclude that $M_i' = M_i \amalg M_i''$. In particular $M' \to M$ becomes étale at a point over p' after an étale base change. This implies that $M' \to M$ is étale at p' (for example by Morphisms, Lemma 0476). We will prove $U' \cong M' \times_M U$ after we prove the claim.

Proof of the claim. Observe that U_p and $U'_{p'}$ have finitely many points. For $u \in U_p$ we have $\kappa(u)/\kappa(p)$ is algebraic, hence $\kappa(u)$ is separably closed. As $U' \to U$ is étale, we conclude the morphism $U'_{p'} \to U_p$ induces isomorphisms on residue field extensions. Let $u' \in U'_{p'}$ with image $u \in U_p$. By assumption (3) the morphism of scheme theoretic fibres $(s')^{-1}(u') \to s^{-1}(u)$, $(t')^{-1}(u') \to t^{-1}(u)$, and $G'_{u'} \to G_u$ are isomorphisms. Observing that $U_p = t(s^{-1}(u))$ (set theoretically) we conclude that the points of $U'_{p'}$ surject onto the points of U_p. Suppose that u'_1 and u'_2 are points of $U'_{p'}$ mapping to the same point u of U_p. Then there exists a point $r' \in R'_{p'}$ with $s'(r') = u'_1$ and $t'(r') = u'_2$. Consider the two towers of fields

$$\kappa(r')/\kappa(u'_1)/\kappa(u)/\kappa(p) \quad \kappa(r')/\kappa(u'_2)/\kappa(u)/\kappa(p)$$

whose "ends" are the same as the two "ends" of the two towers

$$\kappa(r')/\kappa(u'_1)/\kappa(p')/\kappa(p) \quad \kappa(r')/\kappa(u'_2)/\kappa(p')/\kappa(p)$$

These two induce the same maps $\kappa(p') \to \kappa(r')$ as $(U'_{p'}, R'_{p'}, s', t', c')$ is a groupoid over p'. Since $\kappa(u)/\kappa(p)$ is purely inseparable, we conclude that the two induced maps $\kappa(u) \to \kappa(r')$ are the same. Therefore r' maps to a point of the fibre G_u. By assumption (3) we conclude that $r' \in (G')_{u'_1}$. Namely, we may think of G as a closed subscheme of R viewed as a scheme over U via s and use that the base change to U' gives $G' \subset R'$. In particular we have $u'_1 = u'_2$. We conclude that $U'_{p'} \to U_p$ is a bijective map on points inducing isomorphisms on residue fields. It follows that $U'_{p'}$ is a finite set of closed points (Algebra, Lemma 00GA) and hence $U'_{p'}$ is closed in U'. Let $J' \subset A'$ be the radical ideal cutting out $U'_{p'}$ set theoretically.

Second part proof of the claim. Let $\mathfrak{m} \subset C$ be the maximal ideal. Observe that $(A, \mathfrak{m}A)$ is a henselian pair by More on Algebra, Lemma 09XK. Let $J = \sqrt{\mathfrak{m}A}$. Then (A, J) is a henselian pair (More on Algebra, Lemma 09XJ) and the étale ring map $A \to A'$ induces an isomorphism $A/J \to A'/J'$ by our deliberations above. We conclude that $A' = A \times A''$ by More on Algebra, Lemma 09XI. Consider the corresponding disjoint union decomposition $U' = U \amalg U''$. The open $(s')^{-1}(U)$ is the set of points of R' specializing to a point of $R'_{p'}$. Similarly for $(t')^{-1}(U)$. Similarly we have $(s')^{-1}(U'') = (t')^{-1}(U'')$ as this is the set of points which do not specialize to $R'_{p'}$. Hence we obtain a disjoint union decomposition

$$(U', R', s', t', c') = (U, R, s, t, c) \amalg (U'', R'', s'', t'', c'')$$

This immediately gives $M' = M \amalg M''$ and the proof of the claim is complete.

We still have to prove that the canonical map $U' \to M' \times_M U$ is an isomorphism. It is an étale morphism (Morphisms, Lemma 02GW). On the other hand, by base changing to strictly henselian local rings (as in the third paragraph of the proof) and using the bijectivity $U'_{p'} \to U_p$ established in the course of the proof of the claim, we see that $U' \to M' \times_M U$ is universally bijective (some details omitted). However, a universally bijective étale morphism is an isomorphism (Descent, Lemma 02LC) and the proof is complete. □